Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/25/2004
PI Name:	Jacobs, Christopher Rae Ph.D.		
Project Title:	Loading-Induce Fluid Flow Mediates Microgravity Related Disuse Bone Loss		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	94305-3030	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	12/01/2002	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1601		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The bone loss that occurs with disuse in microgravity is a significant challenge that must be overcome in order for long-duration space flights such as a manned mission to Mars to be successful. Recently it has been shown that interstitial fluid flow in bone due to loading is a potent physical signal in regulating bone cell metabolism. The purpose of this project is to determine if this physical signal is also important in the formation of new bone forming cells (osteoblasts). To determine if bone fluid flow due to loading can regulate the formation of new bone cells in terms of proliferation and differentiation of osteoblast precursor cells. We will also determine if the biochemical mechanism involved is similar to the one we have desribed involved in the regulation of mature bone cells. An understanding of the cellular and biochemical mechanism involved in the regulation of bone metabolism due to mechanical loading will result in novel targets for pharmacological intervention, potentially leading to a new class of drugs which could mimic the effects of loading on bone, even in the absence of loading or in the presence of reduced loading.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Disuse bone loss is a critical problem in settings other than microgravity. These include osteoporosis related to age, since inactivity is a known contributing factor, as well as bone loss that occurs with paralysis and stroke.
Task Progress:	We have completed the first aim of the project. Specifically, we have demonstrated that fluid flow due to loading can increase the proliferation of osteoblast precursors as well as increasing their differentiation into mature osteoblasts. These findings have been accepted for publication in the Journal of Orthopaedic Research. We have also made a related discovery (not part of the original project goals) that the dynamics of the time-pattern of loading can have a significant effect on the impact of fluid flow on bone cell metabolism. This finding has been accepted for publication in the Journal of Biomechanics. We are currently working to better define the time course of the effect of flow on gene expression in osteoblast precursors and to determine if the biochemical signaling pathway activated by flow in precursors is the same as in mature osteoblasts.
Bibliography Type:	Description: (Last Updated: 02/08/2013)
Papers from Meeting Proceedings	Batra, N.; Li, Y.J.; Jacobs, C.R. "The effect of short term recovery periods on fluid flow induced calcium oscillations." Orthopaedic Research Society. Feb-2003
Presentation	Li, Y.J.; Batra, N.; Yellowley, C.E.; Jacobs, C.R. "Effect of fluid induced oscillatory shear stress on human marrow stromal cells." Orthopaedic Research Society. Feb-2003
Presentation	You, L.; Li, Y.J.; Batra, N.; Jacobs, C.R. "The effect of oscillatory fluid flow on human marrow stromal cell proliferation." American Society of Mechanical Engineers Summer Bioengineering Meeting. Jun-2003